AP BIOLOGY TOPIC 2.4

Membrane Permeability: Selective Barrier & Cell Wall Support

UNIT 2: CELL STRUCTURE AND FUNCTION (10–13% EXAM WEIGHT)

ENE-2.C | ENE-2.D | SELECTIVE PERMEABILITY | CELL WALLS

Credits: mitali120501@gmail.com

§1. KEY TERMS & FORMAL DEFINITIONS LEXICON — TOPIC 2.4 (MEMBRANE PERMEABILITY: SELECTIVE BARRIER & CELL WALL SUPPORT)

CED Core Vocabulary

Selective Permeability — The plasma membrane regulates the entry and exit of substances, allowing some molecules to cross freely while restricting others based on polarity, charge, and...

Allows — Nonpolar, hydrophobic molecules to dissolve into the lipid core and diffuse rapidly across the membrane without protein assistance.

Passive Transport — Spontaneous movement of molecules down their concentration gradient ([High] → [Low]); ΔG < 0 , requires NO metabolic energy (no ATP) : 1. Simple Diffusion:...

Carrier Proteins (Permeases) — Bind a specific solute (e.g., glucose or amino acids) and undergo a conformational shape change to translocate it across the membrane.

Role of the Hydrophobic Interior Core — Nonpolar fatty acid tails form a hydrophobic core inside the phospholipid bilayer.

Repels / Blocks — Charged ions (surrounded by hydration shells) and large polar molecules cannot penetrate the nonpolar interior on their own → require specific transport proteins .

2. Facilitated Diffusion — Passive movement of polar or charged molecules down their gradient assisted by specific transmembrane proteins: Channel Proteins: Provide a hydrophilic...

Cell Wall Overview — Rigid extracellular layer located outside the plasma membrane in plants, fungi, bacteria, archaea, and many algae. (Absent in animal cells, which have an...

§2. SELECTIVE PERMEABILITY & ROLE OF THE HYDROPHOBIC CORE

The Bouncer Rule

HIERARCHICAL CONCEPT & PATHWAY MAP [CAUSE ➔ MECHANISM ➔ OUTCOME]

Selective Permeability & Role of the Hydrophobic Core ➔ **Selective Permeability** ➔ **Role of the Hydrophobic Interior Core**

- **Selective Permeability:** The plasma membrane regulates the entry and exit of substances, allowing some molecules to cross freely while restricting others based on **polarity, charge, and molecular size**.
- **Role of the Hydrophobic Interior Core:**
  - Nonpolar fatty acid tails form a **hydrophobic core** inside the phospholipid bilayer.
  - **Allows:** **Nonpolar, hydrophobic molecules** to dissolve into the lipid core and diffuse rapidly across the membrane without protein assistance.
  - **Repels / Blocks:** **Charged ions** (surrounded by hydration shells) and **large polar molecules** cannot penetrate the nonpolar interior on their own → require specific **transport proteins**.

[CORE AXIOM / REMEMBER] **Plasma Membrane = Bouncer at the Club (Selective Entry):** "Like dissolves like" — nonpolar passes through the nonpolar core freely; polar/charged need a VIP pass (membrane transport protein)!

[AP EXAM FRQ SCORING STRATEGY] **FRQ Must-Write Phrase:** Whenever explaining why an ion (Na^+, K^+, H^+) or polar molecule (glucose) cannot cross the membrane directly, ALWAYS explicitly state: *"Because the interior core of the phospholipid bilayer is hydrophobic/nonpolar, it repels polar and charged substances."*

§5. PASSIVE TRANSPORT: SIMPLE VS. FACILITATED DIFFUSION

No ATP Required

- **Passive Transport:** Spontaneous movement of molecules **down their concentration gradient** (**[High] → [Low]**); **ΔG < 0** , requires **NO metabolic energy (no ATP)**:
  - **1. Simple Diffusion:** Movement of **small, nonpolar molecules** (O_2, CO_2, N_2 , steroid lipids) directly across the phospholipid bilayer without any transport protein. Rate is directly proportional to solute concentration gradient (linear kinetics).
  - **2. Facilitated Diffusion:** Passive movement of **polar or charged molecules** down their gradient assisted by specific transmembrane proteins:
    - **Channel Proteins:** Provide a hydrophilic tunnel/pore through the membrane core (e.g., **aquaporins** for rapid water transport; gated ion channels for Na^+, K^+, Ca^{2+}, Cl^-).
    - **Carrier Proteins (Permeases):** Bind a specific solute (e.g., **glucose** or amino acids) and undergo a **conformational shape change** to translocate it across the membrane.
- **Why Selective Permeability Matters for Homeostasis:**
  - Maintains distinct internal cytosolic concentrations vs. extracellular environment.
  - Controls **ion & voltage gradients** (membrane potential), selective **nutrient uptake**, toxic **waste removal**, and intracellular pH/osmoregulation.

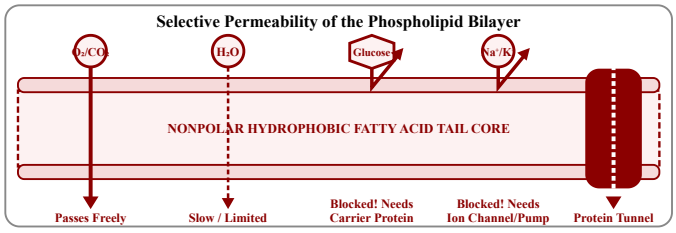
§3. WHAT CAN CROSS THE LIPID BILAYER? (PERMEABILITY HIERARCHY)

High-Yield Comparison

MOLECULE CLASS	EXAMPLES	BILAYER PERMEABILITY	TRANSPORT MECHANISM
1. Small Nonpolar Molecules	O_2, CO_2, N_2	**HIGH (Pass Freely & Rapidly)**; dissolve in hydrophobic tails	**Simple Diffusion** directly through phospholipid bilayer (no protein, no ATP)
2. Hydrophobic / Lipid-Soluble	Steroid hormones (estrogen, testosterone), hydrocarbons, fatty acids	**HIGH (Pass Readily)**; nonpolar rings/chains	**Simple Diffusion** across membrane to bind *intracellular receptors*
3. Small Uncharged Polar Molecules	H_2O, NH_3, glycerol, ethanol, urea	**LOW / LIMITED (Pass Slowly)** in small amounts through transient gaps	Slow simple diffusion + rapid **Facilitated Diffusion** (e.g., H_2O via **aquaporins**)
4. Large Polar Uncharged Molecules	**Glucose** ($C_6H_{12}O_6$), sucrose, amino acids, nucleotides	**IMPERMEABLE (Cannot Pass Directly)**; too large & polar	**Facilitated Diffusion** (GLUT carrier proteins) or **Active Cotransport**
5. Charged Ions (Any Size!)	Na^+, K^+, Ca^{2+}, Cl^-, H^+, HCO_3^-	**VIRTUALLY IMPERMEABLE**; charge + hydration shell blocked by tails	**Ion Channels** (facilitated diffusion) or **ATPase Pumps** (active transport)

§4. MEMBRANE PERMEABILITY SPECTRUM VISUALIZED

Cross-Bilayer Trajectories



§6. CELL WALLS: STRUCTURAL SUPPORT & OSMOTIC PROTECTION

Plants, Fungi & Prokaryotes

HIERARCHICAL CONCEPT & PATHWAY MAP **[CAUSE → MECHANISM → OUTCOME]**

Cell Walls → **Cell Wall Overview** → **Core Functions of the Cell Wall**

- **Cell Wall Overview:** Rigid extracellular layer located **outside the plasma membrane** in plants, fungi, bacteria, archaea, and many algae. *(Absent in animal cells, which have an extracellular matrix [ECM] of collagen/proteoglycans instead!)*
- **Core Functions of the Cell Wall:**
  - **Structural Support & Shape:** Maintains mechanical rigidity and cell shape against gravity.
  - **Prevents Osmotic Lysis (Bursting):** When water enters by osmosis in a hypotonic environment, the rigid wall pushes back to create **turgor pressure** ($+\Psi p$) rather than bursting.
  - **Permeability Barrier Note:** Cell walls are **porous and freely permeable** to water, O_2/CO_2, and small dissolved solutes — the **plasma membrane underneath** is what actually performs selective permeability!

TAXONOMIC GROUP	DIAGNOSTIC CELL WALL POLYMER	CHEMICAL MONOMER / STRUCTURE
Plants & Green Algae	**Cellulose** (+ pectin & lignin in wood)	Unbranched **β-1,4-glucose** polysaccharide microfibrils; perforated by **plasmodesmata**
Fungi (Yeasts, Molds)	**Chitin**	Nitrogen-containing polysaccharide (**N-acetylglucosamine**); same polymer as arthropod exoskeletons!
Bacteria (Eubacteria)	**Peptidoglycan**	Modified sugar chains cross-linked by short polypeptides (targeted by **penicillin & lysozyme**)
Archaea	**Pseudopeptidoglycan** / S-layer proteins / polysaccharides	**Lacks true peptidoglycan** (insensitive to penicillin)

[AP EXAM FRQ SCORING STRATEGY] **Polymer Identification on the AP Exam:**

- Cell wall with **Nitrogen (N)** in a eukaryotic decomposer → **Chitin (Fungi)**
- Cell wall with **only C, H, O** (**β-glucose**) in a photoautotroph → **Cellulose (Plants)**
- Cell wall with **sugars + amino acids** in a prokaryote → **Peptidoglycan (Bacteria)**

[COMMON AP EXAM FRQ TRAP] **Common Trap:** Do NOT say the cell wall is "selectively permeable" like the plasma membrane! The cell wall is a relatively coarse, porous meshwork (like a rigid basket); **selective permeability is strictly a property of the phospholipid plasma membrane**.

§7. SCHOLARLY SYNTHESIS: EXPERIMENTAL DESIGN, QUANTITATIVE ANALYSIS, PATHOLOGY & AP FRQ RHETORIC

High-Yield Exam Rigor

MASTER TOPIC SYNTHESIS & CAUSAL ARCHITECTURE — TOPIC 2.4 **[MOLECULAR CAUSE ➔ CELLULAR MECHANISM ➔ PHYSIOLOGICAL / ECOLOGICAL EFFECT ➔ FRQ JUSTIFICATION]**

Topic 2.4: Membrane Permeability: Selective Barrier & Cell Wall Support ➔ **Structural / Energetic Determinants** ➔ **Regulated Pathway & Homeostasis** ➔ **Experimental Perturbation (±2 SE_x̄)** ➔ **Phenotypic / Evolutionary Outcome**

SCHOLARLY DIMENSION	COMPREHENSIVE ACADEMIC REFERENCE & MECHANISTIC SPECIFICATION
I. Experimental Design & Required Lab Controls	AP Investigation 4 (Diffusion & Osmosis): Dialysis tubing & potato core mass change (% ΔMass = [(Final − Initial) / Initial] × 100) across sucrose molarities (0.0 M to 1.0 M). IV: Extracellular sucrose molarity (M); DV: % change in mass; Isotonic point: x-intercept where % ΔMass = 0% (no net water movement). **Null Hypothesis (H_0):** State explicitly that the independent variable treatment has *no statistically significant effect* on the dependent variable; any observed difference is due to random sampling chance alone.
II. Quantitative Formulas & Graph Interpretation	Quantitative Formulas & Graph Rules: Total Water Potential: $\Psi = \Psi_p + \Psi_s$. Solute Potential: $\Psi_s = -iCRT$ (where i = ionization constant [1 for sucrose/glucose, 2 for NaCl], C = molarity in mol/L, R = 0.0831 L·bar/mol·K, T = K = °C + 273). Surface-Area-to-Volume Ratio (SA:V): Cube SA = $6s^2$, V = s^3 (SA:V = 6/s). Larger SA:V accelerates diffusion per unit volume. **Error Bar Rule (±2 SE_x̄):** If the ±2 SE_x̄ error bars of two treatment groups *do not overlap*, there is a statistically significant difference between means ($p < 0.05$; reject H_0).
III. Pathology, Mutation & Disruption Case Study	Clinical & Physiological Disruption Case Study: Cystic Fibrosis results from a deletion (ΔF508) in the CFTR chloride channel protein, preventing misfolded channels from reaching the apical plasma membrane. Impaired Cl^- efflux prevents osmotic water movement into mucus, producing thick, dehydrated mucus in lungs and pancreatic ducts. Ouabain/Digoxin inhibits the Na^+/K^+-ATPase pump, collapsing the electrochemical gradient required for secondary active cotransport.
IV. AP FRQ Task Verb Scoring Templates	• **DESCRIBE / IDENTIFY:** State the direct trend, value, or structural component with exact units from the figure. • **EXPLAIN (How/Why):** Connect cause to effect via a 3-link chain: *"[Change in X] causes [molecular/cellular mechanism Y], which results in [observable outcome Z]."* • **PREDICT:** State the specific directional change (increase, decrease, or remain constant) when the pathway is perturbed. • **JUSTIFY:** Pair quantitative data/evidence from the prompt with the underlying biological principle from Topic 2.4 to prove why the claim holds.

Credits: mitali120501@gmail.com

Support this site
Scan to donate